Question 1 - Dictionaries and For Loops

You are given a dictionary where **each key is a student name (string)**, and the corresponding **value is a list of their exam scores (integers)**:

```
grades = {
    "Alice": [88, 92, 79],
    "Bob": [95, 85, 91],
    "Charlie": [70, 68, 75],
    "Diana": [100, 100, 100]
}
```

Write a Python script that:

1. Iterates over the dictionary using a for loop.

2. For each student, it computes: (1) the average score, (2) whether the student passed or failed (average $\geq$ 80 → pass, else fail)

3. Creates a new dictionary called results where **the key is the student name** and **the value is a tuple of (average_score, "Pass"/"Fail")**

4. Prints the results dictionary at the end.

```
{
    'Alice': (86.33, 'Pass'),
    'Bob': (90.33, 'Pass'),
    'Charlie': (71.0, 'Fail'),
    'Diana': (100.0, 'Pass')
}
```

Question 2 - Conditional Logic

You are allowed to use **only one `elif()`**.Write a Python script that does the following:

1. Prompt the user (or assign a variable in the code) for an **integer x**.

2. Check the value of **x** and print a message according to these rules:

- If **x is negative**, print **"x is negative"**. If **x** is positive, print **"x is positive"**.

- If **x is zero**, print **"x is zero"**.

- If **x is negative and even**, print **"x is negative and even"**.

- If **x is positive and odd**, print **"x is positive and odd"**.

Question 3 - List Processing with Functions

a. Write a Python function called **process_numbers(numbers)** that takes a list of integers as input and performs the following tasks:

1. Filter out negative numbers from the list.

2. Compute the square of each remaining number.

3. Return a new list containing the squared numbers in the same order.

Then, pass this list to the function you defined: **nums = [3, -1, 0, 5, -7, 2]**

Store the result in a variable called squared_list and print it: **[9, 0, 25, 4]**

b. Implement the same function using a list comprehension following the syntax below:

```
def process_numbers(numbers):
   return [your list comprehension]
```

Question 4 - NumPy

Use np.set_printoptions(precision=2)

Vector Creation

Create a vector called vector1 containing the integers **0 through 14** (inclusive) **without using np.array().**

- Set the random seed to **123**, then create a second vector called vector2 of the same length using the **np.random** module.
- Compute and print the following three operations between the vectors:
  a. The **element-wise sum**
  b. The **dot product**
  c. The **element-wise multiplication**

Matrix Creation

Create a two-dimensional NumPy array of size **2 × 3** containing only ones **without using np.array()**.

- Print the **shape** (2, 3) **and dimension** (3) of the array.
- Compute the **dot product** of this matrix A with its transpose A^T
- Print the resulting matrix and print its shape.

Reshaping and Concateniating

Use the vector1 and vector2 from the first step:

- Reshape both vectors into **column vectors (14 × 1).**
- Concatenate the two column vectors **horizontally** (i.e., along **axis=1**) to form a **15 × 2** array.